DEPARTMENT OF COMPUTER AND INFORMATION SYSTEMS ENGINEERING
BACHELORS IN COMPUTER SYSTEMS ENGINEERING
Course Code and Title: CS-323 Artificial Intelligence
Complex Engineering Problem
TE Batch 2024, Fall Semester 2026
TERM PROJECT
Grading Rubric

Group Members	**Student No.**	**Name**	**Roll No.**	**Student No.**	**Name**	**Roll No.**
	S1			**S3**		
	S2			**S4**		

CRITERIA AND SCALES				**Marks Obtained**			
				S1	**S2**	**S3**	**S4**
Criterion 1: Formulation and implementation of graph-based search methods. (CPA-1, CPA-2, CPA-3) **[4 marks]**							
1	2	3	4				
The algorithm is incorrectly implemented, produces wrong outputs, or fails to execute on test boards.	The algorithm is partially implemented, produces limited or inconsistent results.	The algorithm is correctly implemented and produces mostly correct outputs.	The algorithm is correctly and efficiently implemented.				
Criterion 2: Formulation and implementation of constraint satisfaction problem (CSP). (CPA-1, CPA-2, CPA-3) **[4 marks]**							
1	2	3	4				
The CSP formulation or backtracking implementation is incorrect or incomplete.	The CSP model is partially correct but produces limited or inaccurate outputs.	The CSP formulation is correct, producing mostly accurate results with acceptable efficiency.	The CSP solver is accurately and efficiently implemented.				
Criterion 3: Integration of scheduling and navigation (CPA-2, CPA-3) **[2 marks]**							
1	2	3	4				
CSP and search components are not properly integrated, or the robot fails to execute the cleaning task.	Integration is partially functional but fails on important steps, such as visiting all locations or returning to the charging station.	The integrated system executes the schedule and completes the journey in most valid cases, with minor issues.	Both components are correctly integrated; the robot follows the generated schedule, cleans all required locations, returns to the charging station.				
Criterion 4: Comparison, evaluation, and analysis of both methods (CPA-3) **[2 marks]**							
1	2	3	4				
No meaningful comparison or analysis; results not discussed or evaluated.	Basic comparison made but lacks depth, clarity, or quantitative support.	Clear comparison with relevant metrics and reasonable interpretation.	Comprehensive and insightful comparison using quantitative and qualitative metrics.				
Criterion 5: Adherence of report to the given format and requirements. **[4 marks]**							
1	2	3	4				
The report does not contain the required information and is formatted poorly.	The report contains the required information only partially but is formatted well.	The report contains all the required information but is formatted poorly.	The report contains all the required information and completely adheres to the given format.				
Criterion 6: Individual and team contribution. (CPA-2) **[4 marks]**							
1	2	3	4				
The student did not contribute meaningfully to project tasks.	The student contributed partially to assigned tasks with limited collaboration.	The student contributed adequately and met assigned goals.	The student demonstrated active participation, initiative, and significant contribution beyond expectations.				

Final Score = _____ Teacher's Signature: _____